



ISRO discovers non-Earth bacteria

Indian space scientists have discovered three new strains of bacteria, not found on Earth, in the upper stratosphere

ESA's Mars Express has found signs of erosion, allowing geologists to further study the history of the red planet. The Mars Express has also beamed back video footage of water on the surface of Mars.

SCI-TECH

advantage if they were not answered. The US stayed close on the heels of every space achievement of the USSR. The US sent a man into space barely 23 days after Russia sent hers. However, the US could not match all the early Soviet achievements. This included sending the first woman into space, the first multi-man crew, the first probe to another astro-body (the moon) and the first photos from the far side of the moon. The US, was however well ahead in terms of communication satellite launches, and orbital technologies but these were technicalities that the population of the countries did not generally appreciate. For everyone, the Soviets were simply more sophisticated because they were the first to explore into territory where no one else had gone before.

In 1961, John F Kennedy, then President of the United States, famously declared that the US would attempt a mission of "landing a man on the moon and returning him safely to the earth". The Soviets removed themselves from this race early in the decade, and did not even try. This declaration, and the events that followed, marked a fundamental change in the nature of human activity in space. Instead of gaining a tactical military advantage over an enemy, the Apollo missions were about exploring the unknown, for the sake of scientific knowledge. The space race slowly died down, and was replaced by a more co-operative environment.

The next big thing after the US Apollo missions was the MIR space station of the Soviets. The name "MIR" means "peace" in Russian. The MIR space station was a forerunner to the International Space Station, using the same methods of modular fabrication in space. The US and the Russians finally collaborated on a famous 1995 shuttle mission, when the US space shuttle Atlantis docked with the Russian space station. Several more docking visits followed.

The adventure of space has finally brought a number of nations into cooperation — a fundamental requirement because of the economics involved. Co-operative efforts turn out to be more cost effective than independent efforts, and a shared pool of ideas and efforts lets everyone achieve more. The Americans, the Russians, the Chinese and the EU are working together on the ISS. The EU has also collaborated with the Chinese and Indians on launches.

Fundamentally, each space mission is made up of two components: The first component is the launch vehicle, which is the rocket that goes into space. The second component is the payload, which can be an artificial satellite, a probe, or a spacecraft. The primary Russian launch vehicles are the Soyuz and the Proton rockets. The Saturn V took the Apollo astronauts to the Moon. NASA currently uses a fleet of launch vehicles — the most famous being Taurus.

India, too, has a fleet of launch vehicles, most of which are intended for satellite launches. These are SLV, PSLV and GSLV, and simply stand for Satellite Launch Vehicle, Polar Satellite Launch Vehicle and Geosynchronous Satellite Launch Vehicle, respectively. The Chinese have sent spacecraft with humans into orbit with one of their Long March rockets, which is an umbrella term for their fleet. The Japanese and the EU too have launch vehicles, but these are for satellite payloads. NASA's Space Shuttles are the most well-known space craft in existence. Columbia and Challenger were both lost in accidents, while Atlantis, Endeavor and Discovery are still in service. The Russians currently use the Soyuz craft; the program is in its third generation of models. China used the Shenzhou craft, which successfully carried three people into space in 2003. The Indians can launch men into space using their PSLV rockets.

Particularly noticeable in the last decade is co-operation. Payloads from one

The Soyuz spacecraft



The Soyuz spacecraft is the most successful space flight model in history. Both the launch vehicle and the spacecraft are named Soyuz, and there have been many modifications over the years. Although the shuttles came in and are being phased out, the Soyuz space craft has been around and still continues.

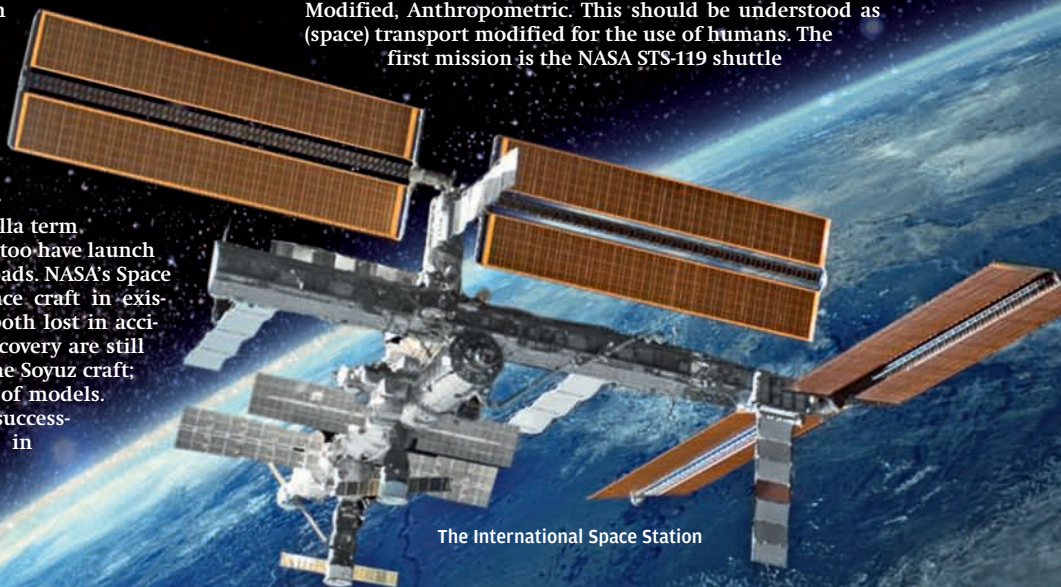
Each Soyuz craft consists of three parts. The first part is a spherical orbital module, which provides life support and accommodation for the crew during the mission. The descent module is bell-shaped, and returns the cosmonauts back to the Earth, by deploying three parachutes on re-entry. The instrumentation module has the engine and the solar panels for energy. Only the descent module is shielded for re-entry, the other two modules are destroyed in the stratosphere. The Shenzhou space craft of China and India's Orbital Vehicle are both based on the Russian Soyuz. In fact, Russia has shared design plans with those two countries.

country have gone into space using launch vehicles from another — Italian satellites have gone up using ISRO's launch vehicles, for example. Countries without any launch capabilities have sent astronauts to space on other space crafts. Even modules within a spacecraft of one country are designed by another — such as the Italian multi-purpose logistics modules on American space shuttles for cargo. The usually reserved Russians have given space technology to the Koreans. The private sector is expected to be involved soon, when the shuttles are phased out.

Although there is widespread co-operation, there are a few independent missions being undertaken by various countries, and most have a personal agenda. This agenda is not competitive, but an internal effort to gain technical know-how. There are exciting times ahead. The next decade will see many more players, culminating in a climax in 2020 where four agencies may meet on the moon.

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This year will see a number of human spaceflights. Current spaceflights from the US are denoted with the letters STS (for Space Transportation System) followed by a number. The Russian missions are referred to as TMA, which stands for Transport, Modified, Anthropometric. This should be understood as (space) transport modified for the use of humans. The first mission is the NASA STS-119 shuttle



The International Space Station